

We sometimes face constrained optimization problems:

(9)

$$\min_x f(x)$$

$$\text{s.t. } g_i(x) \leq 0 \text{ for all } i=1, \dots, m$$

this form is known as ^{the} standard form, all other constrained optimization problem can be formulated as it.

For example,

$\max_x f(x)$ can be transformed into it by considering equivalently $\min_x -f(x)$ equation constraint

Can be transformed $g(x) = 0 \rightarrow g(x) \leq 0$
 $\rightarrow -g(x) \leq 0$

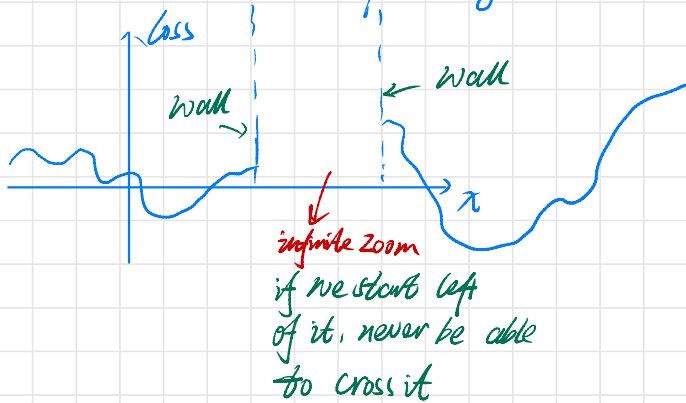
A simple (but might not be practical) way to solve (9) is to convert it into a unconstrained problem by considering the penalty method.

$$\min_x \left(f(x) + \sum_{i=1}^n I(g_i(x)) \right) \rightarrow \text{add penalty here}$$

$$\text{where } I(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ +\infty & \text{otherwise} \end{cases}$$

$\rightarrow$ or a super large value

(this could make the loss space very bad)



Lagrange multiplier method:

Let us consider the Lagrange form:

$$L(x, \lambda) = f(x) + \lambda^T g(x), \text{ where } \lambda = (\lambda_1, \dots, \lambda_n)^T \\ g(x) = (g_1(x), \dots, g_n(x))^T$$

Then, we have the original problem is **equivalent** to the following problem:

$$\min_x \max_{\lambda} L(x, \lambda) \\ \text{s.t. } \lambda \geq 0$$

note that for any $\varphi(x, y)$, we have:

$$\min_x \varphi(x, y) \leq \varphi(x, y) \leq \max_y \varphi(x, y) \\ \Rightarrow \max_y \min_x \varphi(x, y) \leq \min_x \max_y \varphi(x, y)$$

Hence, we have the solution for the following

Lagrange Dual problem:

$$(d) \quad \max_{\lambda} \min_x L(x, \lambda) \\ \text{s.t. } \lambda \geq 0$$

will be smaller or equal to the original problem (p).

Let us compare the original problem (p) and dual problem (d).

Denote $q(\lambda) = \min_x L(x, \lambda)$ (this unconstrained problem for λ)
is known as inner problem

Say we can solve it (often possible in analytical form)

We can see the dual problem (d) is much easier than the original problem (p).

(1) The constraints goes from

$$g_i(x) \leq 0 \rightarrow \lambda_i \geq 0$$

(Could be very complex)

→ this is known as "box constraints"
the easiest constraints

(2) $\min_x f(x)$ could be a non-convex problem

$\max_{\lambda} q(\lambda)$ is a convex problem for λ .

(3) Dimensionality reduction.

Say $x \in \mathbb{R}^{10,000}$ and we have 5 constraints, then $\min_x f(x)$ could be very hard

By considering $\max_{\lambda} q(\lambda)$, we reduce the dimension into 5.

In short, the dual problem could be dramatically simpler than the original problem.

However, keep in mind that the results for dual problem (d) could be smaller than the original problem (p)

The difference between the results for original and dual problem is

duality gap: $G = \underbrace{q^*}_{\text{results for original problem}} - \underbrace{d^*}_{\text{results for dual problem}}$

if $G = 0 \rightarrow$ Strong duality

$G > 0 \rightarrow$ Weak duality (this means we have a lower bound for the original problem, which implies the x found in the dual problem is not in the feasible region of the original problem)

When will Strong Duality hold?

Is necessary but not sufficient Condition is known as

Slater's Condition:

1. Convex: $f(x)$ and $g_i(x)$ are convex
2. Feasibility: There is at least one x such that $g_i(x) < 0$ for all $i=1, \dots, n$
 $\downarrow$
strictly smaller

$\downarrow$ this could happen because in the inner problem

$$\min_x (f(x) + \lambda^T g(x))$$

$\downarrow$ when violate $g_i(x) < 0$, get penalty $\lambda_i g_i(x)$.

however if $f(x)$ is very deep (i.e. very small) this violation might be worth it...

While the Lagrange method gives us a often simpler dual problem, it does not solve the problem directly.

We often solve a constrained optimization directly using:

KKT Conditions

→ gives us a local minimum, if Slater's condition holds, a global minimum

1. Stationarity

$$\nabla_{\lambda} f(x) + \lambda^T \nabla g(x) = 0$$

inner constraint problem
 $\min_x L(x, \lambda)$

if x minimize it, the gradient must be 0.

→ Lagrange form

2. Primal Feasibility

$$g_i(x) \leq 0 \quad \text{for } i=1, \dots, n$$

3. Dual Feasibility

$$\lambda_i \geq 0, \quad \text{for } i=1, \dots, n$$

4. Complementary Slackness

$$\lambda_i g_i(x) = 0$$

→ if $g_i(x) < 0$, then $\lambda_i = 0$
if however $g_i(x) = 0$, then

Recall that the original problem is equivalent to

$$\min_x \max_{\lambda \geq 0} L(x, \lambda)$$

$$= \min_x \max_{\lambda \geq 0} (f(x) + \lambda^T g(x))$$

1. Stationarity

if local minimum, then the gradient must be 0 (for every x_i)

2. Complementary slackness

if $g_i(x) < 0$, then $\lambda_i = 0$

because for fixed x , λ maximize $L(x, \lambda)$.

Steps for using KKT Conditions:

(Sufficient condition)

KKT Conditions are a checklist whether x^*, λ^* we have in hand result valid local (or global if Slater's condition holds) results.

But it can also be used to find the local minimum. The idea is to construct x^*, λ^* to satisfy KKT Conditions.

Steps:

1. Guess which points are on the boundaries (following complementary slackness)

2. Solve x and λ from the stationarity and complementary slackness
 $\nearrow$ say k_1 $\downarrow$ k_1 equations $\downarrow$ k_2 equations

3. Check feasibility, if pass, find a local minimum,
if not, start from step 1 again.

note that different guess in step 1 could give us different local minimum. In general, there are total 2^{k_2} different combinations.
 $\downarrow$
number of constraints